

Allergy Relief in Humid Climates

Dust mites are a problem in new homes in humid climates. What keeps them away?

by Subrato Chandra and David Beal



The increasing incidence of allergy and asthma, particularly among children, is a worldwide concern. About 20% of the general population suffer from allergies, and about 5% suffer from asthma. Among children the percentage of allergy sufferers is even higher. Since most of us stay indoors more than 90% of the time, the air quality in buildings is of paramount importance in maintaining respiratory health.

Energy-efficient buildings with tight building envelopes may make for poor indoor air quality—yet allergy-resistant houses must be airtight, well insulated, and well ventilated. One cannot control and filter the ventilation air unless the house is airtight, nor can one maintain a positive pressure. Cold spots and cold floors may be present in houses that are poorly insulated. High local relative humidity (RH) hastens the formation of molds on these cold surfaces and is conducive to the breeding and proliferation of dust mites. So energy-efficient construction is a prerequisite for fighting dust mites, but it is not sufficient in and of itself.

Allergens and Irritants

The major allergens found in homes today are environmental tobacco smoke

Table 1. Characteristics of Houses Tested

LOCATION	ORLANDO				JACKSONVILLE		NEW ORLEANS		HUNTSVILLE	
House	New 1	Old 1	New 2	Old 2	New 3	Old 3	New 4	Old 4	New 5	Old 5
Constructed	12/95	6/95	10/96	8/91	11/96	5/94	10/96	~1900	12/96	~1960
Area (ft ²)	3,746	3,746	3,003	2,002	2,368	2,949	3,897	5,996	2,885	1,905
RH Control	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
w/w Carpet	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Occupants	4	4	4	3	2	3	3	8-10	4	2
Pets	0	2 Cats	0	0	1 Cat	2 Cats	0	0	0	0
Vacuum Type	Upright	Central	Upright	Upright	Upright	Upright	Central	Upright	Central	Upright
Dust Bag	Regular	–	AT*	AT	Regular	AT	–	Regular	–	Regular
Vacuum Frequency	1/Week (wk.)	1-2/wk.	1-2/wk.	1-2/Month	2/wk.	2-3/wk.	1/wk.	2-5/wk.	1/wk.	1-2/wk.

*allergen trapping

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and dust mite allergens, the focus of this article. Other house allergen sources and irritants include pollen, pet dander, fungi and molds, cockroaches, endotoxins (from bacteria), volatile organic compounds, fine mineral dust, and radon. (For more general information on fighting allergens, see “Creating Air-tight and Healthy Homes,” *HE* Nov/Dec '01, p. 18.)

Environmental tobacco smoke.

About 25% of North American adults smoke today. Sidestream and exhaled smoke (environmental tobacco smoke, or ETS) is a major source of breathable particles indoors—and of various carcinogens, other toxic compounds, and odors as well. Some air-filtering and -purifying devices can clear up ETS somewhat. High levels of localized ventilation may also help. However, the most practical solution seems to be banishing the smokers to the outside of the house. Building science is probably of limited help in alleviating ETS.

Dust mites. In humid climates, dust mites are a leading cause of asthma. In the hot, humid central Florida climate, approximately 60% of patients complaining of respiratory problems tested positive to a dust mite skin prick test—rates 2 to 3 times greater than the rates for any other allergen. In a study of North

Carolina schoolchildren, 30% tested positive for house dust mite allergy.

Dust mites feed on human skin flakes and are about 300 micrometers (mm) in size. There are two types of dust mite allergens. Type 1 allergens, the most common form, come from dust mite fecal particles. Type II allergens come from dust mite body parts. The fecal particles are 5–10 mm in size and heavier than air. When airborne, they settle in about five minutes. Dust mites live and breed on carpets (see “Which Carpets Do Dust Mites Like Best?” p. 32), sofas, beds, pillows, and other soft and fluffy furnishings. A person lying down on a bed or walking across a carpet can release puffs of dust containing dust mite fecal particles. These particles are easily breathed in before they can be filtered out using a room or whole-house air filter. Dust mite allergen concentrations of 2 micrograms (µg) per gram of dust or higher can become a health problem for house occupants.

Current recommended actions to reduce allergen exposure in the home include encasing mattresses, pillows, and box springs; washing bedding weekly in water hotter than 130°F; and the regular use of vacuum cleaners that have allergen trapping features or that are ducted to the outside (central vacuums). Another

current recommendation is to remove carpets from houses in humid climates.

Laboratory studies of dust mites indicate that their bodies will dehydrate in an environment with RH below 58% at a temperature of 77°F, or RH below 52% at 59°F. While dehydration does not necessarily lead to death, it reduces fecal and allergen production. Along with thorough vacuuming, dehumidification can help to control dust mites and dust mite allergens.

A Study of New and Existing Homes

Since many consumers living in humid climates prefer carpeting in their new houses, we performed a study to determine the best way to eliminate dust mite allergens in new houses with wall-to-wall carpets and in humid climates. We studied five pairs of homes, all with central cooling and heating systems (see Table 1).

The homes were in four cities in the southeastern United States (Orlando, Florida; Jacksonville, Florida; New Orleans, Louisiana; and Huntsville, Alabama). Each home pair consisted of a new home with a central dehumidifier and an older existing home without a central dehumidifier.

Which Carpets Do Dust Mites Like Best?

Wall-to-wall carpets seem to be popular with home buyers. So along with knowing the aesthetic, insulating, and soundproofing qualities of each kind of carpet, it would be helpful for builders, contractors, building scientists, and occupants to know which kinds of carpet are hospitable to dust mites—and which aren't.

Glenda Gilmore Andes, a graduate student at Virginia Polytechnic Institute and State University, and her colleagues did a pilot study of three common carpet materials to find out which one dust mites prefer. The three materials tested were nylon; wool (treated with permethrin, a chemical that protects the wool from

moths and beetles); and olefin. Nylon is the most popular of these materials; it is used in 65% of carpets.

Eight samples of each material were used in the study. Samples were all multilevel loop carpet with a low pile. Each sample was laced with yeast (food for the dust mites), infested with 50 American house dust mites, and placed in an environment controlled for temperature (70°F–77°F) and humidity (50%–87% RH) for four or six weeks. The samples were then removed and the dust mites were extracted and counted.

The average number of mites in the nylon carpet samples was 122.

The average for the permethrin-treated wool samples was 57, and the average for the olefin carpet samples was 58. The study showed that nylon, the most common carpet material used, was the material most hospitable to dust mites.

Please note that the homes used in the dust mite allergen prevention study had nylon carpeting, yet they were allergen free because of thorough vacuuming and central dehumidification. Also note that nylon carpeting is a low emitter of volatile organic compounds (per tests conducted by Al Hodgson at Lawrence Berkeley National Laboratories).

The existing homes were located in the same city as the new homes and were used for comparison purposes.

Each home was monitored on average for 12 months (minimum 8 months, maximum 16 months). Interior temperatures and RH were measured at the carpet level by small battery-powered data loggers, which recorded the instantaneous data every hour. Dust samples were collected in each home at a minimum of four locations—master bedroom mattress, master bedroom carpet in front of the bed, family room sofa, and the family room carpet in front of the sofa.

Dust mite allergens in the air are hard to measure as they quickly settle down. Since mites live near surfaces, we sampled surface dust monthly by vacuuming an area of approximately 11 ft² for one minute. Dust samples were collected with a portable vacuum cleaner whose inlet was modified to accept a standard paper coffee filter, which collected the dust. Students collected dust samples in all the Orlando area homes. Homeowners, who were given a similar set of dust collection equipment and instructed in its use, collected dust samples in the rest of the homes.

Existing homes. The summertime average temperatures for the five existing homes were below or near 77°F

The average RH for the existing homes during the summer months (second and third quarters of every year) was below or near 58%. During the winter months (first and fourth quarters), the central air conditioners, which dehumidify indoor air, were not used much. Due to occupant activities and infiltration of outside humid air, the average carpet level RH for all the existing homes during the winter quarters was higher than it was during the summer quarters, while the average carpet level temperatures went down during the winter. This promotes dust mite growth in the winter months in the southeastern United States.

As expected, the allergen levels of the existing homes were high, with one exception—house 3 (see Figure 1). For the purpose of our study, we defined thorough vacuuming as vacuuming at least twice a week with an allergen-trapping dust bag, or once a week with a central vacuum cleaner exhausted to the outside of the conditioned living area. Existing home 1 had a central vacuum cleaner, which was used regularly. Despite not having a central dehumidifier, this house had low allergens for the first three quarters of the year. It was only in the last quarter of sampling, after a period of prolonged high humidity,

that the maximum allergen level in the living room carpet exceeded the threshold value of 2 µg per gram of dust.

Existing home 2 had an allergen-trapping vacuum cleaner, but it was not used often. Consequently, the house had high allergen levels. Existing houses 4 and 5 had standard vacuum cleaners, and, despite frequent vacuuming, had significant allergen levels. In existing home 3, a housekeeper vacuumed two or three times a week with an allergen-trapping vacuum cleaner. This house never had measurable allergens.

New homes. The yearly average measured RH values for the new homes were lower than 58%, although some days the RH did exceed 58%. The central dehumidification systems performed well in controlling the winter indoor humidity. The average temperatures were comparable to those in the existing homes.

New home 1 received a regular cleaning with a standard vacuum and, despite RH control, had a significant number of dust mite allergens (see Figure 2). We discovered studying data from new home 1 that new homes could be infested with dust mites in a few weeks. This is also reported in the literature. The fact that this new house had high allergen levels is also consistent with the

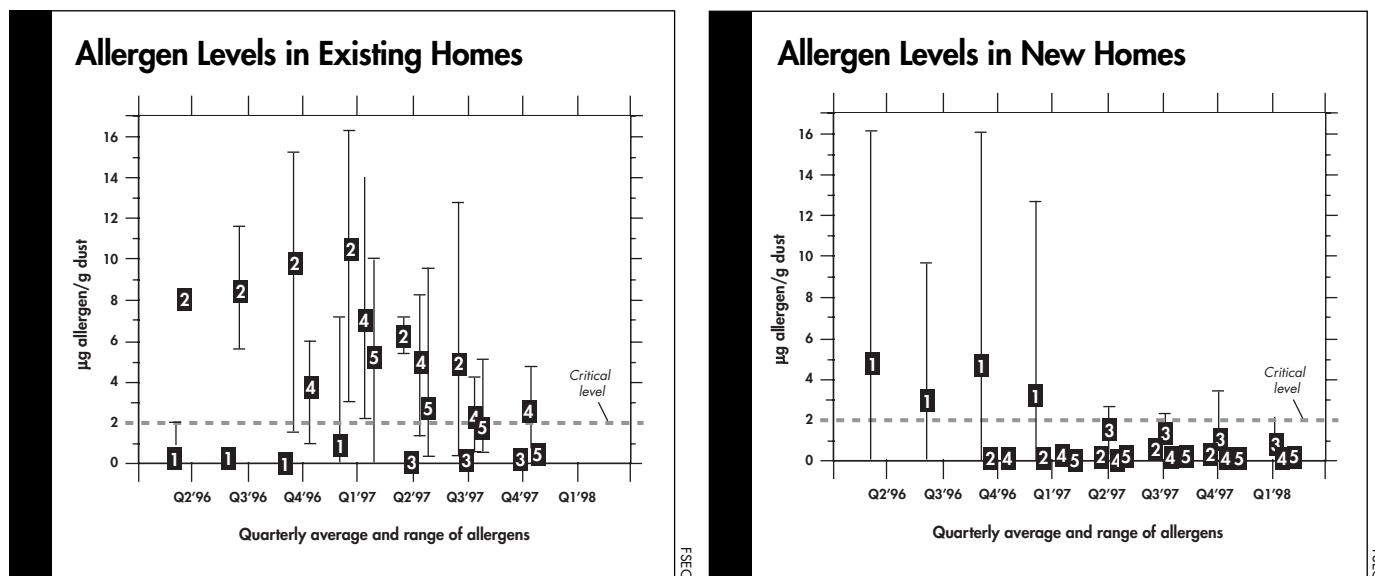


Figure 1. (left) Measured allergen levels were high in the existing homes we studied, except for house 3, which was thoroughly vacuumed twice a week. **Figure 2.** (right) The results of allergen measurements in new house 1, which was vacuumed with a standard vacuum cleaner and had humidity control, show that humidity control alone is not sufficient to keep dust mite numbers below a critical level.

results of other studies that found that RH control alone was ineffective in controlling dust mite allergens.

New homes 2, 4, and 5 all had wall-to-wall carpets and they all received thorough vacuum cleaning. The highest measured type 1 allergen level at any location in these three homes was only 1.04 µg per gram of dust. Although new home 3 did not have wall-to-wall carpeting, just area rugs, it had allergen levels occasionally exceeding the threshold value. The homeowners used a standard vacuum cleaner.

How to Maintain an Allergen-Resistant House

Washing bed sheets and clothing in hot water (at least 130° F) and covering mattresses, box springs, and pillows with impermeable covers can reduce exposure to dust mite allergens, as explained above. So can removing carpets from homes in humid climates. But our study showed that, even in humid climates, thorough vacuuming and a central dehumidification system installed in conjunction with central cooling and heating can maintain dust mite allergens below clinically significant levels in new homes with wall-to-wall carpeting.

Whole-house dehumidification that runs independently of the central heating and cooling is recommended to control mold, mildew, endotoxins, and dust mites. Independent whole-house dehumidification is especially important in the months when the weather is humid, the temperature is mild, and the cooling or heating system does not operate frequently. Room dehumidifiers are unlikely to control humidity throughout the house, although they might benefit single rooms.

Our research showed that it is difficult to maintain RH below 58% at the floor level in humid climates at all times, even if the average air RH level is maintained at or below that level with whole-house central dehumidifiers. However, we still consider this to be a very promising strategy, especially when combined with thorough vacuum cleaning.

Looking to the Future

The new houses studied (average floor area is 3,180 ft²) are larger than typical new U.S. homes (average floor area is 2,225 ft²). Census data indicates that in 1997, 15% of new home construction consisted of houses with 3,000 ft² or more of conditioned area. The number of occupants of new houses in our study did

not exceed four per house. Future studies should look at smaller new and existing homes with higher occupant densities, lower-cost central dehumidification systems, and allergen trapping-vacuum cleaners to determine the range of housing types that can benefit from this promising dust mite control strategy. Future studies should also quantify the benefit of thorough vacuum cleaning alone in controlling dust mite allergens.



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